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Numerical Investigation of Lateral Jet with Supersonic Reacting Flow

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To improve the understanding of the lateral jet, this paper reports the computational fluid dynamics simulations of a cone–cylinder–flare model with lateral jet control, especially with regard to the interaction of a cold- or hot-gas jet with a supersonic crossflow. The Spalart–Allmaras turbulence model is used to solve the three-dimensional compressible Navier–Stokes equations, and the methodology is first validated by comparing the numerical results with the wind-tunnel data. The predicted axial pressure coefficient profiles are acceptably accurate with the experimental data. However, in the case of a hot-gas jet without considering the chemical reaction, there is a significant error at the separation point compared with the experimental result. Using a simplified reaction mechanism, the chemical reaction is carried out in the hot-gas jet simulation. The numerical results show that the separation of the boundary layer takes place at a distance of eight times of the nozzle diameter, which is very close to the experimental result. The aforementioned results indicate that, in the simulation of lateral jet interference, it is necessary to consider the chemical reaction, which can improve the numerical accuracy of predicting both the aerodynamics performance and flow structures.

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Nomenclature

C_p	=	pressure coefficient
D	=	nozzle diameter, mm
M	=	Mach number
PR	=	jet total to freestream static pressure ratio; P_{0J}/P_∞
P_0	=	freestream total pressure, bar
P_{0J}	=	jet total pressure, bar
P_∞	=	freestream static pressure, bar
p	=	pressure, bar
T	=	temperature, K
T_0	=	freestream total temperature, K
T_{0JC}	=	total temperature of cold jet, K
T_{0JH}	=	total temperature of hot jet, K
T_∞	=	freestream static temperature, K
γ_C	=	specific heat ratio of cold jet
γ_H	=	specific heat ratio of hot jet
ΔC_p	=	differential pressure coefficient

I. Introduction

TO KEEP the high motility in a rarefied gas, vehicles usually employ reaction control systems (RCSs), which can provide the thrust required for the attitude control of supersonic aircrafts, launch vehicles, and spacecrafts. Compared to aerodynamic control surfaces, the RCS has the advantages of high control efficiency and short response time in the rarefied gas at high altitude. In addition, unlike aerodynamic control surfaces, the RCS does not increase drag when it is not in operation [1,2]. However, in a supersonic flow environment, with strong interference between the supersonic jet and crossflow, ejection of the gas along the lateral direction intensely changes the pressure distribution around the body and produces additional aerodynamic force and torque on the vehicles. Consequently, the interference between the lateral jet and crossflow can cause crucial consequences on in-flight behavior. Therefore, aerodynamic effects related to the RCS have attracted a wide range of research interests in the last decades [3].

In recent years, several research groups have made great progress in the study of the flowfield of lateral jet interference. One series of experimental data appropriate for the validation of computational fluid dynamics (CFD) simulations was recently published [4–6]. These studies involve wind-tunnel experiments on a missile model with side jet control, and they report the detailed wall pressure measurements for a number of jet configurations with crossflow conditions. In the experimental study of the jet temperature, it is found that the high-temperature jet enlarges the upstream high-pressure zone and diminishes the area of the lower-pressure zone. Therefore, the additional positive thrust of the high-temperature jet is larger than those of the low-temperature jets. With the increasing of the jet pressure ratio, the aforementioned phenomena are more significant [6]. In addition, many scholars have done sufficient research on these two models by means of numerical simulations [7–9]. These results indicate that the Mach number, jet pressure, and nozzle number have significant influence on the lateral jet interference. The amplification factors of force and torque are increased with the increasing of Mach number [10]; the bow shock and the separation region grow with the increasing of the jet pressure ratio or the number of nozzles [11].

With the development of CFD and computer hardware, more and more scholars have adopted complex turbulence models to simulate lateral jets [1,12,13]. DeSpirito [9,14] studied the effects of various turbulence models on the jet interaction and observed that, of all the turbulence models investigated, the Menter models gave the best results. Watanabe et al. [15] performed a large-eddy simulation for a transverse jet in a Mach 1.9 crossflow to investigate the effect of injectant species on the turbulent structure and the mixing of the jet. The results show that the hydrogen (H_2) and helium (He) jets have a much higher velocity difference between the jet and crossflow in the near field, and the observed turbulent intensities for these two injectants are much higher than those for nitrogen (N_2) and ethylene (C_2H_4). It is worth noticing that, regardless of the hot-gas jet simulation or experiments, it is assumed that the propellant gas injected from the nozzle is chemically inert and does not cause chemical reaction with crossflow under the high-temperature condition [7].

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However, relevant studies indicate that this hypothesis might have a negative impact on the numerical results. Li et al. [16–18] considered the chemical nonequilibrium flow method and simulated the hot rocket staging. To reduce the computational complexity, a two-dimensional model was used to simulate the hot staging. The results illustrate that, compared to the chemically nonequilibrium flow simulation (which is of higher thermochemical level), there is 36% underestimation on the length of the separation zone when using a single-species model and 10% underestimation when using a two-species model.

In the present study, numerical simulations are carried out using an in-house code on a typical cone–cylinder–flare (CCF) configuration to study the surface pressure distribution and aerodynamic forces around the model. By comparing the numerical results with the archival experimental data, we are emphatically analyzing the difference of the pressure distribution between the nonreaction results and the chemical nonequilibrium flow numerical results. This paper is organized as follows: The methodology is mainly presented in Sec. II. In the next section, the authors adopt the same flow conditions as the experimental parameters of the DLR, German Aerospace Center's (DLR) CCF missile model to simulate the cold-gas jet and hot-gas jet, we verify the methodology with the archival experimental data, and we demonstrate the accuracy through a grid-independence test. In Sec. IV, using the hot gas containing $H_2/CO/CO_2$ and O_2, the authors carry out the numerical simulation of the lateral jet by a chemical nonequilibrium flow method with the simplified chemical reaction mechanism. We compare the results of differential surface pressure coefficients and flowfield structures. Through the aforementioned results, we study the influence of chemical nonequilibrium flow on the interaction between the lateral jet and crossflow. Finally, the paper ends with the conclusions.

II. Methodology

A. Meshes for Configurations

The computational domain used for the model is the same as reported by Gnemmi et al. [19]. Only one-half of the CCF model is shown in Fig. 1, which consists of a $2.8D$-long conical nose, a $3.2D$-long cylindrical section, and a $3D$-long flared afterbody with a $1.66D$ base diameter. The jet nozzle is circular, and the sonic nozzle of diameter $d_j = 4.6$ mm ($0.05D$) is located on the cylindrical section at the position of $4.3D$ from the nose and the circumferential angle $\varphi = 180$ deg. In the domain, the x axis is taken along the longitudinal direction from the nose of the model, whereas the y and z axes are along the normal and lateral directions. As the crossflow is supersonic, the inflow domain is taken $1D$ ahead of the nose tip and the downstream section ends at the end of the flare section of the model. In the freestream version, the far-field boundary at the jet injection zone is taken at distance of $5D$ from the model axis. A hexahedral mesh is used for discretizing the computation domain, and the mesh refinement is performed in the region near the jet nozzle and the cylinder. The spacing of the first layer of cells is set as 1×10^{-6} m from the model surface in order to reach a y^+ value near one. Supersonic boundary conditions corresponding to crossflow Mach number of three and the attack of angle of 0 deg is prescribed at the inflow plane, and sonic condition is assumed to be at the injector exit plane. As the outflow boundary is supersonic, flow variables are extrapolated from the interior values in the outlet boundary.

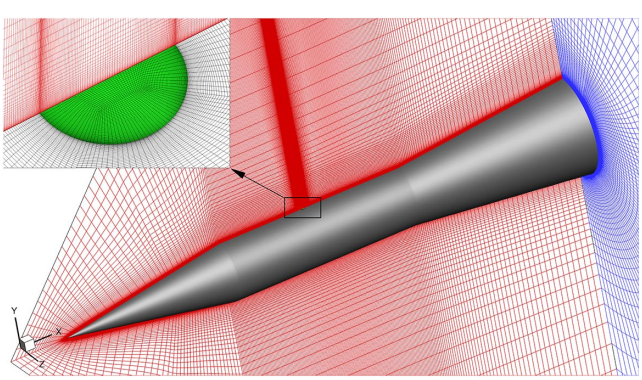

Fig. 1 CCF model with hexahedral grid.

B. Flow Conditions

The flow conditions are shown in Table 1. The crossflow is simulated with mass fraction of 21% of O_2 and 79% of N_2. The cold gas adopts the same gaseous species as the crossflow. Similar to Höld et al. [8] and Despirito [14], the hot gas in the present study is consisted of only five of the seven products listed, eliminating the copper and lead products. The mass fractions are 38.21% of CO_2, 1.73% of H_2, 10.47% of H_2O, 35.47% of CO, and 14.12% of N_2. For the hot-gas simulation, it is assumed that the gas mixture is chemically inert. The jet parameters can be calculated by isentropic relation. In the cold-gas simulation, $\gamma_C = 1.399$ and the jet exit temperature is $T_{JC} = 244$ K. In the hot-gas simulation, $\gamma_H = 1.233$ and the jet exit temperature is $T_{JC} = 2058$ K.

C. Computational Fluid Dynamics Solver

The governing equations for the three-dimensional Reynolds-averaged Navier–Stokes (RANS) simulations are solved by an in-house code [20,21]. We develop the unstructured code of cell-centered finite volume method, which is successfully applied to solving many scientific and engineering applications [22,23]. By using Strang temporal splitting [24], the code can also solve the nonequilibrium reacting flow [25,26]. Zhang et al. [27] used this chemical nonequilibrium uncoupled algorithm to simulate the flow in a scramjet nozzle. The simulation results showed that the radical recombination processes influenced by inflow conditions and nozzle scales are consistent with Damköhler numbers and potential dissociation energy release. The RANS equations are modeled on the hexahedral grid using the Spalart–Allmaras (SA) turbulence model [28]. In the present numerical simulations, the Van Leer–Hänel upwind flux scheme [29] is employed for discretizing the inviscid flux vectors, and the viscous flux vectors are discretized by a central-difference scheme. The implicit lower–upper symmetric Gauss–Seidel method [30] is employed for time marching of flow equations [31]. To enhance the solution stability, the chemical reaction equations are solved by the variable-coefficient ordinary differential equation [32] and α quasi-steady-state [33] methods.

III. Numerical Results and Comparison with Measurements

Using the DLR's CCF missile model, the experimental data and numerical results are more abundant. By summarizing the results of the literature, the model contains typical flowfield structures [34], such as the bow shock and the λ shock in front of the injection and the barrel shock around the jet and the Mach disk, which are all indicated in Fig. 2. The horseshoe vortices that are generated upstream of the injection and wrapped around the side jet can be clearly seen. In this work, these flowfield structures are expected to be reproduced in the numerical simulations. Firstly, the simulation of the "no-jet" condition and the static pressure of 0.923 bar for crossflow is carried out to obtain the reference result. Similar to the experimental measurements [6], the surface pressure is extracted along the model surface of $\varphi = 180$ deg and $2.8D \leq x \leq 6D$ ($D = 90$ mm). Secondly, different "jet-on" conditions are simulated using the jet boundary conditions listed in Table 1. Because these investigations focus on the interaction process of the jet and crossflow, the differential pressure coefficients and the typical flowfield structures between the cold-gas jet and hot-gas jet are analyzed and discussed in the following paragraphs.

A. Grid Refinement

In this section, the grid convergence is the primary concern. To match the size of the experimental model, the diameter D of the model is modified from 90 to 40 mm and $d_j = 4.0$ mm, and the

Table 1 Computational conditions for DLR

Parameter	Value
Crossflow	
M	3.0
P_0, bar	34
T_0, K	293
P_∞, bar	0.923
T_∞, K	105
PR	130
Cold-gas jet	
P_{0J}, bar	120
T_{0JC}, K	293
Hot-gas jet	
P_{0J}, bar	120
T_{0JH}	2300

proportion of the computational model remains the same as described in Sec. II.A. The numerical simulations adopt the same flow conditions as in literature [9,19]. The Mach number is 2.8, and the PR is 100. Three different grids of 1.39, 2.10, and 3.13 million cells are employed for the grid refinement test. Figure 3 shows the variation in the pressure coefficient along the model of $\varphi = 180$ deg. The x scale in the figure is nondimensionalized. The numerical results show agreement with the experimental data. In addition, the grid convergence index described by Roache [35,36] is also computed. In the present simulation, the surface pressure has been taken as the parameter of interest, and the other parameters can refer to the literature [37]. The results can be observed in Fig. 3, in which the grid convergence index (GCI) is seen to be less than 10% for the pressure. Thus, the solutions may be considered to be grid independent. Furthermore, the difference between the results of the medium and fine meshes is less significant compared with that between the coarse and medium meshes. Hence, the medium grid consisting of 2.1 million cells is chosen as a reasonable tradeoff between the computational cost and numerical accuracy.

B. Cold-Gas Jet

The cold-gas lateral jet uses the air as the jet flow. The flowfield structures are shown in Figs. 4 and 5. It can be observed that the bow shock induced by the transverse momentum of the jet interacts with the boundary layer ahead of the jet nozzle. The adverse pressure gradient induces separation, which in turn yields a family of compression waves in front of the jet. The incoming turbulent boundary layer interacts with a shock wave and separates from the model surface, forming a large turbulent shear layer and a recirculation region. Upon the orifice, the jet tries to penetrate and expand into the low-pressure crossflow, and a high-speed shear layer is formed between the ambient air and the jet. Furthermore, as the expansion fan generated at the edges of the injector interacts with the shear layers, the boundaries of the jet are deflected inward, and compression waves from these shear-layer deflections form the barrel shock. Finally, a Mach disk normal to the jet flow compresses the jet fluid. It is worth noting that the barrel shock and Mach disk are unsteady. The unsteadiness of the barrel shock appears to be related to acoustic waves propagating across the jet nozzle [38], and the unsteadiness of the Mach disk is such that a significant portion of the high-velocity jet fluid bypasses the upstream edge of the Mach disk during certain instants [39].

The most dominant features of the interaction flowfield structures are usually the counter-rotating vortex pair (CVP). As mentioned in the literature [40,41], the typical vortical structures of the jet in crossflow include 1) the near-field jet shear-layer vortices; 2) the downstream wake vortices; 3) the horseshoe vortices wrapping around the jet column; and 4) the counter-rotating vortex pair in far field. The vortical structures are

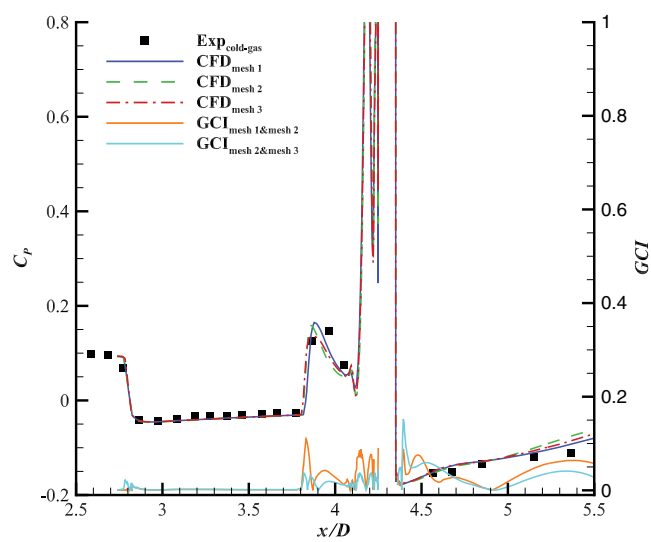

Fig. 3 C_p contours on the model surfaces and grid convergence index.

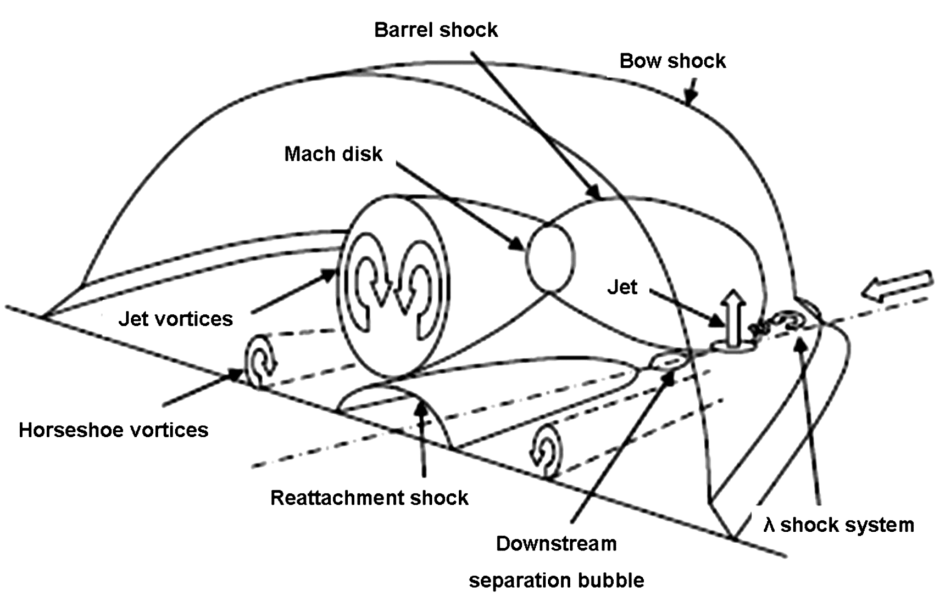

Fig. 2 Schematic of the jet interaction flowfield [3].

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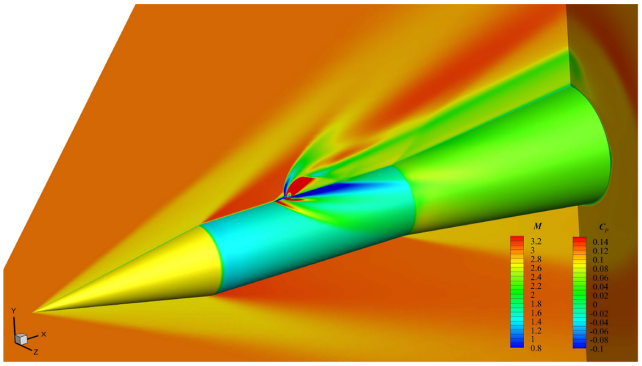

Fig. 4 Typical lateral jet interaction flowfield around the projectile: Mach contours on symmetry and outlet and C_p contours on model surface ($PR = 130$).

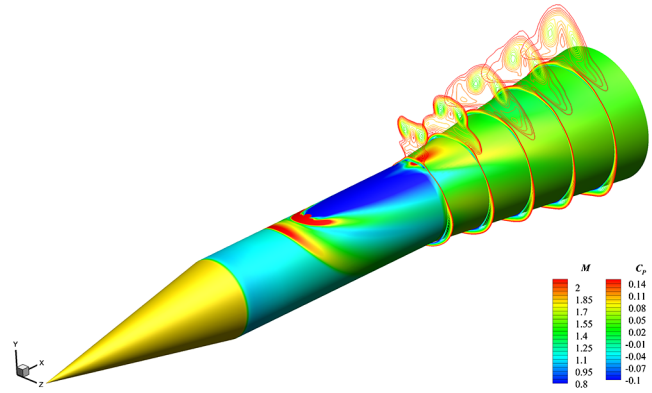

Fig. 6 C_p contours on the model surface and Mach contours on axial slice planes.

shown in Figs. 6 and 7. Horseshoe vortices are formed upstream of the jet and travel downstream. These vortices form because the crossflow boundary layer encounters an adverse pressure gradient upstream of the jet, separates, and forms spanwise vortices that move around the jet. Due to the pressure gradient in the streamwise direction, the jet bends in the direction of the crossflow and forms a CVP that travels downstream. Moreover, the crossflow accelerates around the jet, and a saddle node is observed on the symmetry plane.

Figure 8 shows the differential pressure coefficient ($\Delta C_p = C_{p,\text{jet}} - C_{p,\text{no-jet}}$) distribution measured and computed at the intersection of the symmetry plane of the model surface. Before the position of $x \leq 4.8D$, the numerical results of the cold-gas jet show agreement with the experimental result [6] (plotted as symbols). In the upstream of the jet exit, the pressure increase due to the separation zone and the bow shock is well recognizable. In the downstream, a low-pressure region is produced by the interaction of the jet and crossflow. The comparison results show that there is a large prediction error behind the nozzle, which is about 25% different with the experimental data. The reason causing the error may be that the grid density affects the simulation accuracy. Meanwhile, the Reynolds average method of the RANS model reduces the algorithm accuracy and causes the error in the pressure recirculation zone, which is slightly overestimated.

C. Hot-Gas Jet

The hot-gas jet is considered to be an ideal gas with a constant specific heat [7,14]. However, using the chemical non-equilibrium flow simulation can obtain more consistent results

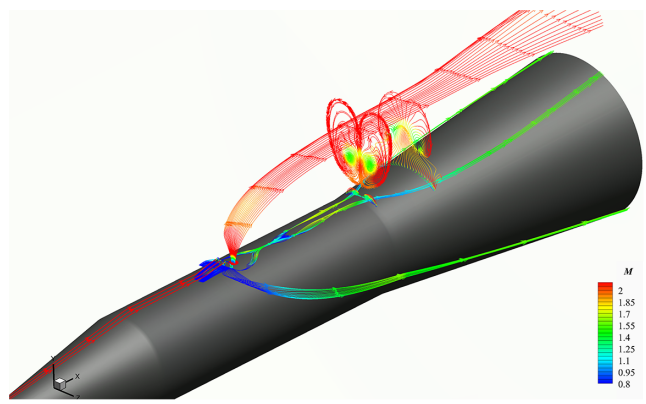

Fig. 7 Model surface with three-dimensional streamlines.

comparing with the experimental data [8]. In fact, as the hot-gas jet tries to penetrate and expand into the crossflow, there is a species diffusion phenomenon near the nozzle. In this section, the simulation of the hot-gas jet without chemical reaction is introduced. Unlike the cold gas, the hot gas uses the approximate combustion product species, and the lead and copper are replaced by other gases. Meanwhile, the total jet temperature has been modified from 293 K (T_{0JC}) to 2300 K (T_{0JH}).

Figure 9 shows the differential pressure coefficient ΔC_p distribution measured and computed at the intersection of the symmetry plane of the model surface. Compared with the cold-gas jet

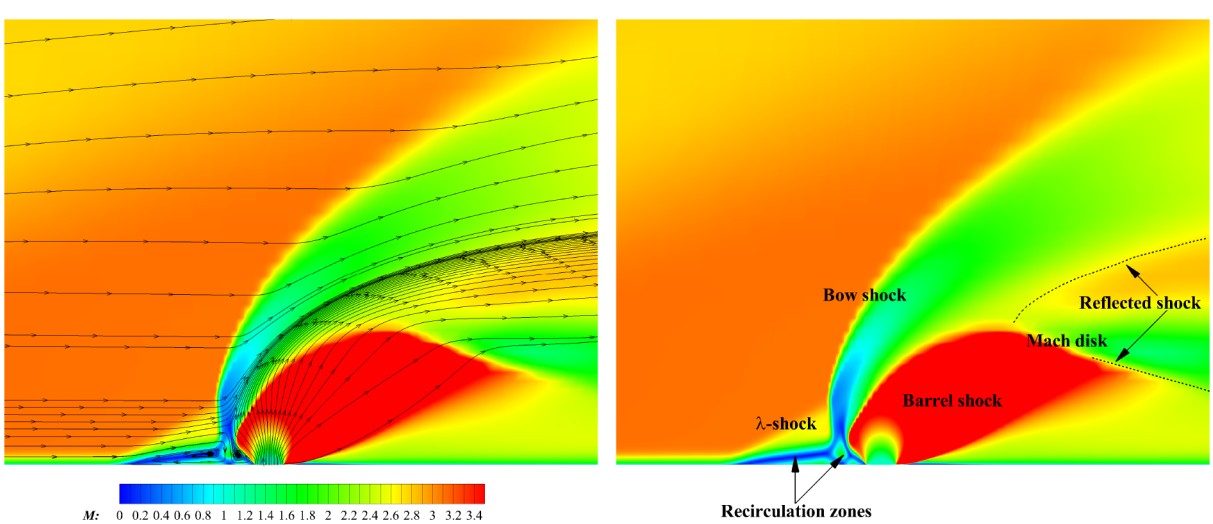

Fig. 5 Mach number contours and flowfield structure on the symmetry plane with streamlines.

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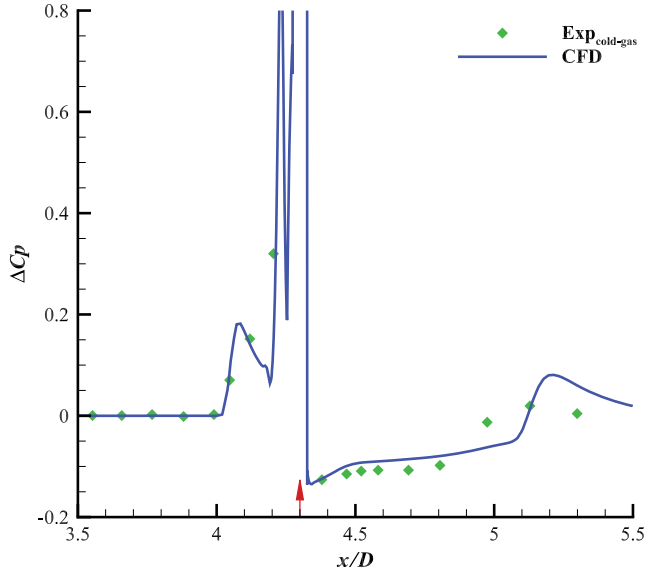

Fig. 8 Differential pressure coefficient distribution on the model surface (cold-gas jet).

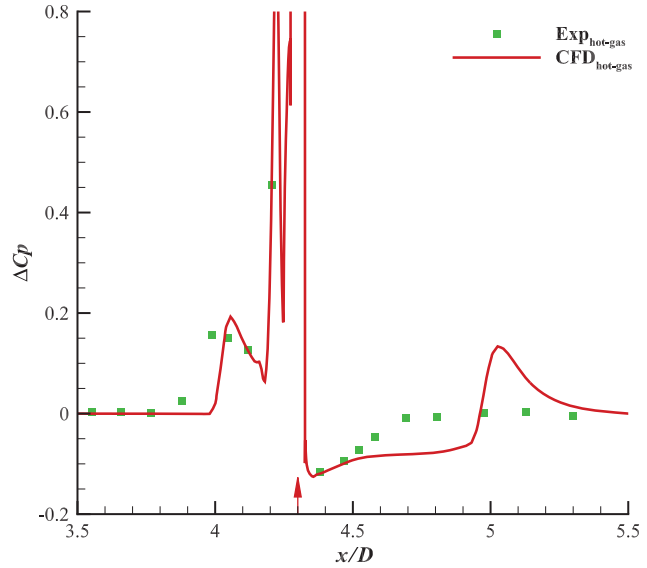

Fig. 9 Differential pressure coefficient distribution on the model surface (hot-gas jet).

simulation, the numerical results of the hot-gas jet simulation are reasonable comparing with the experimental result [6] (plotted as symbols). However, there are some errors in certain numerical values. The plateau pressure of the separation zone captured by CFD simulation is slightly higher than the experimental value, and the separation length is of approximately 50% of the experimental result. In addition, the pressure recovery in the region behind the jet is not predicted accurately. The present results predict an overshoot as the pressure recovers from the low pressure induced by the underexpanded jet, whereas the experiment shows a gradual pressure rise during this process with no overshoot. The same error also arises in the cold-gas jet simulation.

IV. Hot-Gas Jet with Chemical Reaction

In the previous sections, the chemically inert gas is used to simulate a lateral jet under the specific pressure ratio. According to the numerical results, the position of the separation point caused by the reverse gradient does not agree with the experimental result, and the numerical results in other literature [7,14] cannot match the data. Firstly, the errors are quantitatively analyzed. The multispecies jet undergoes chemical reaction with the crossflow at the interface under the high-temperature and high-pressure conditions near the nozzle. The chemical reaction can release more energy, the pressure after the bow shock is further increased, and the higher reverse gradient drives the separation zone to move forward along the model surface. In addition, in similar numerical simulations, Li [16] found that the presence of chemical reactions has a great significance on the separation length in the hot interstage separation case. Considering the chemical reactions, the separation length on the surface of the second stage is larger than that of the single-component gas. Based on the aforementioned understanding, the effect of chemical reactions on the flowfield structures in the hot-gas jet simulation is taken into account.

Chemical nonequilibrium simulations have been widely used in many engineering problems [42,43]. To reduce the computational cost, the chemical reaction mechanism [44,45] is simplified according to the composition ratio. Because the jet contains 35.25% of CO, the crossflow contains 21% of O_2 and the mass fraction of H_2 is negligible; and only the chemical reactions involving CO and O_2 (O atoms) are taken into account as the elementary reaction. The reactions ignore the effects related to H_2 and H_2O as follows:

$$\mathrm{CO} + \mathrm{O} + \mathrm{M} = \mathrm{CO_2} + \mathrm{M} \tag{1}$$

$$\mathrm{CO} + \mathrm{O_2} = \mathrm{CO_2} + \mathrm{O} \tag{2}$$

$$\mathrm{O_2} + \mathrm{M} = \mathrm{O} + \mathrm{O} + \mathrm{M} \tag{3}$$

where M represents a third body element.

Figure 10 shows the comparison of the differential pressure coefficient distribution with the experimental data; the figure also contains the results in Sec. III.C. Considering the chemical reactions, the separation length in front of the nozzle is obviously longer. The separation of the boundary layer takes place at about eight times of nozzle diameters upstream, and the results are very close to the experimental results [5,6]. However, the pressure recovery in the region behind the jet is not predicted accurately. The CFD predicts an overshoot as the pressure recovers from the low pressure induced by the underexpanded jet, whereas the experiment shows a gradual pressure rise during this process with no overshoot. Under the high-temperature and high-pressure conditions, O_2 in the vicinity of the

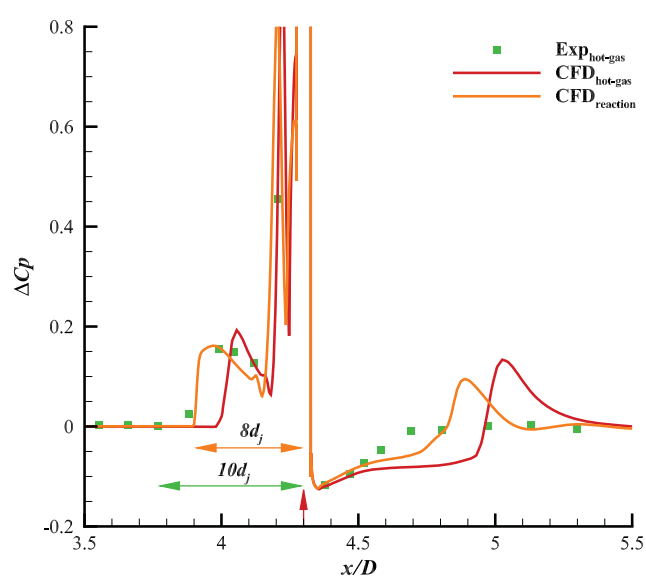

Fig. 10 Differential pressure coefficient distribution on the model surface (hot-gas jet with reaction).

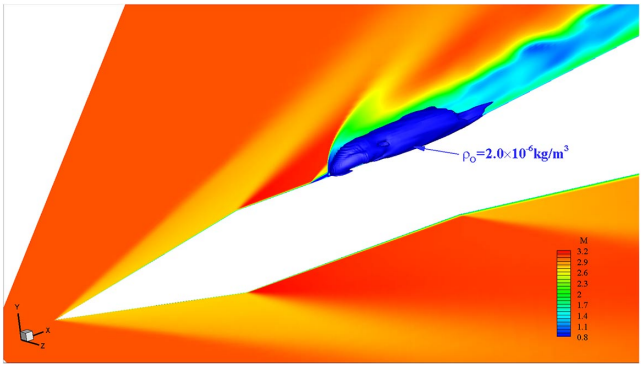

Fig. 11 Isosurface distribution of O and Mach contours on the symmetry plane.

contact surface with the hot gas occurs dissociation reaction, and thus O is produced in chemical reaction (3). Then, O reacts with CO to produce CO_2 in chemical reaction (1). The preceding processes are the positive reaction process. The chemical energy released during the reaction process is the major cause of increasing the separation length, and this phenomenon indicates that the presented views are reasonable. In addition, the effect of the lateral jet on the flowfield structures can be demonstrated by the species distribution. Figure 11 shows the isosurface distribution of O (darkest area in the figure). It can be seen that the O generated by the dissociation reaction is located between the bow shock and the barrel shock. Then, O bends in the direction of the crossflow and completely surrounds the barrel shock.

Figure 12 shows different species density distributions. It can be seen that the density of O increases gradually behind the bow shock, and the maximum value is 2.5×10^{-5} kg/m^3. The density of O outside the barrel shock is higher than that inside the barrel shock, indicating that the chemical reactions are more intense outside the barrel shock. Because the jet does not contain O_2, the density of O_2 is maintained at zero within the nozzle range. O is generated by chemical reaction (1), and chemical reaction (3) does not work. The density distribution of CO_2 is accompanied with the movement of the separation zone. CO_2 appears at the separation point, and the density of CO is almost zero. This phenomenon indicates that the chemical reaction process consumes a lot of CO and increases the concentration of CO_2. Mach number contours on the symmetry plane are shown in Fig. 13. The original flowfield structures such as the λ-shock wave and bow shock wave are maintained, but the scale of the separation zone becomes larger and there are some fluctuations in the flowfield downstream of the nozzle. The possible reason is that

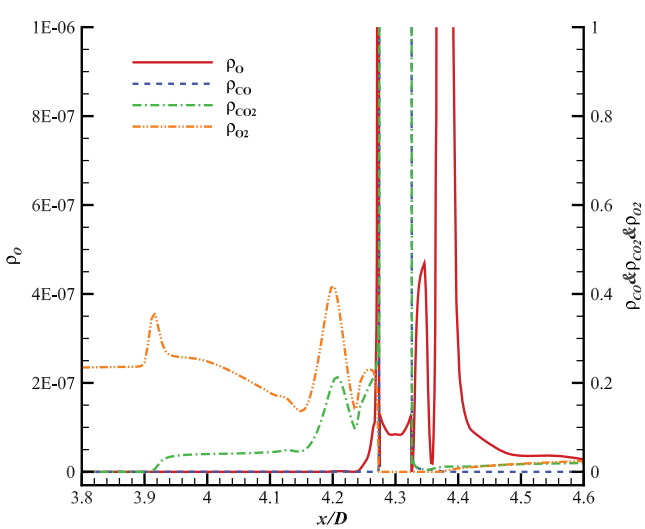

Fig. 12 Density distribution along the central line.

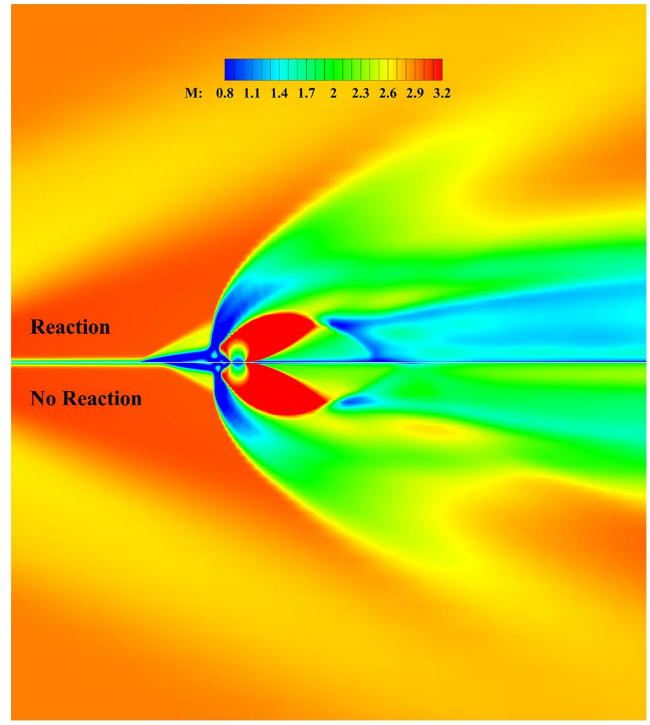

Fig. 13 Mach contours on the symmetry plane.

the combustion phenomena with the finite rate chemical reaction model are not sufficient to cause instability of the flowfield.

V. Conclusions

In this paper, the lateral jet interaction flowfield in supersonic flow is simulated by using the Spalart–Allmaras turbulence model to solve the three-dimensional compressible Reynolds-averaged Navier–Stokes equations. Under the condition of the chemically inert gas, the comparison of the complex wave structures and the axial pressure distribution with the experimental results is carried out. The numerical results are reasonable comparing with the experimental results and validate the accuracy and reliability of the present methodology.

The differential pressure coefficient distribution upstream of the nozzle is consistent with the experimental results in the cold-gas jet simulation, and the plateau pressure of the separation zone can be predicted accurately in the hot-gas jet simulation. However, there is large error in the separation length with the experimental data. In addition, the pressure recovery in the region behind the jet is not predicted accurately. The present results predict an overshoot as the pressure recovers from the low pressures induced by the underexpanded jet, whereas the experiment shows a gradual pressure rise during this process with no overshoot.

By using the simplified chemical reaction mechanism and the chemical nonequilibrium flow method, the numerical results of the hot-gas jet obviously improved. The separation length is eight times of nozzle diameters, which is very close to the experimental results. Meanwhile, the flowfield structures also undergo fundamental changes. The chemical reaction is more intense outside the barrel shock, and O generated by the dissociation reaction completely surrounded the barrel shock. The aforementioned results show that, in the simulation of lateral jet interference, it is necessary to consider the chemical reaction, which could further improve the accuracy of the numerical simulation and be helpful for predicting the aerodynamic characteristics of the aircraft.

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